

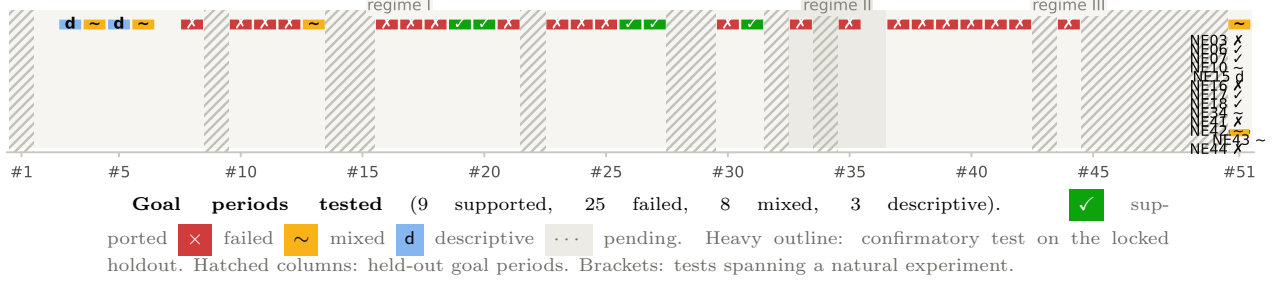
H39: Catalysts vs. fields

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. No operator lever is a pure catalyst: nudges both unstick and tilt agents, while human messages, mentions, erasures and goals act mainly as fields, on activity or on content depending on the regime.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Which operator levers change *what* agents do (fields: stationary occupancy shifts) and which only speed them up (catalysts: escape at fixed occupancy)? Matched-window kinetic decomposition of behavior-state chains around nudges, human messages, @-mentions and context erasures in 32 non-holdout periods, plus step levers. **Round 1b** used the nudge's leading @ as its only target, DQ8's presence-cut lever design, and Jev v3.1 behavior states as a second state space (V4). **Nudges are both**, now in two state spaces: B4 $K = +0.13$, idle share -0.048 (CI now excludes 0); V4 $K = +0.25$, waiting share -0.08 , waiting escape $\times 1.6$. Human messages are fields; @-mentions steer content ($+0.33$ SD), not behavior. Forced erasure is a field toward work in both state spaces (its K flips sign between them). Natives: switching the nudger off is a catalytic loss in both state spaces (NE43); nudges are not more catalytic under the 12-h pause default (NE44). Holdout not run.

PHYSICAL MODEL

Models 02/10: a kinetic Potts (Markov-jump) agent over states $s \in \{\text{work, chat, idle, consolidate}\}$ with Arrhenius rates k_{ij} , well depths G_i and symmetric barriers B_{ij} . A field h tilts the wells; a catalyst c lowers the barriers. Measured per lever: kicked and control transition matrices T^K, T^C in matched 30-min windows, their stationary distributions π^K, π^C , the escape probability $e_s = 1 - T_{ss}$, the field index ϕ , the catalytic index K , and the catalytic fraction ρ .

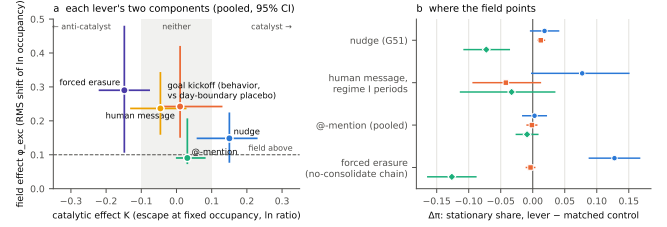
$$k_{ij} = \nu e^{-(B_{ij}-G_i)/T}, \quad \pi_i \propto e^{-G_i/T}$$

$$s_{ij} = \frac{1}{2}(\delta \ln k_{ij} + \delta \ln k_{ji}) \quad (\text{catalytic part})$$

$$a_{ij} = \frac{1}{2}(\delta \ln k_{ij} - \delta \ln k_{ji}) \quad (\text{field part})$$

$$\phi = \text{SD}_{\pi^C} \left[\ln \pi_s^K - \ln \pi_s^C \right], \quad \rho = |K| / (|K| + \phi_{\text{exc}})$$

$$K = \ln \left(\sum_s \pi_s^C e_s^K / \sum_s \pi_s^C e_s^C \right)$$



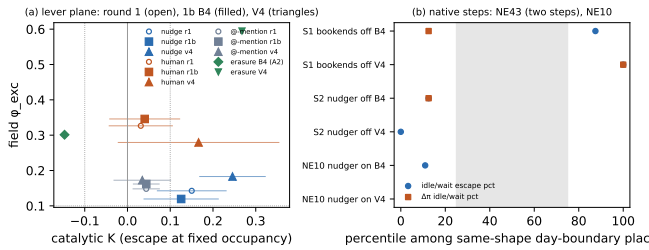
Observables. (a) Pooled field strength ϕ_{exc} (placebo-corrected) against catalytic effect K , with 95% CIs. Dashed line and grey band: the 0.10 thresholds. Only the nudge reaches the “both” corner; human messages and erasure are fields, and erasure is anti-catalytic. (b) Where each field points: stationary share changes for work, chat and idle.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful. An operator gets a concrete rule. To unstick an idle agent, send one directed nudge: escape rises $\sim 50\%$, and the agent also stays out of idle while it acts. To steer what an agent writes without changing how busy it is, mention it. To change topics, set a goal; that works in the large regime-III village. The physics label is softer than it sounds: the field/catalyst split depends on a convention, the nudge verdict rests on one period (G51), and the effects are individual and diluted. Switching the nudger off (08-21, post hoc) barely moved the swarm. “Messages always carry a field” is the robust, slightly deflating lesson. Prompt tweaks, room changes and most fixes showed nothing at day scale. Unconfirmed on the holdout.

RESULTS VS PREDICTIONS

Prediction	Round 1 $\rightarrow$ 1b (B4; V4)	Verdict
P1 nudges: both; K 0.15 $\rightarrow$ 0.13; $\Delta\pi_{\text{idle}}$ -0.043 4/5 B4; both, idle down (n.s.) $\rightarrow -0.048^*$; V4 both, K 0.25*, 5/5 V4 $\Delta\pi_{\text{wait}}$ -0.08^*		
P2 human: field (P 0.92); chat -0.03 ; V4 $\times$ both, toward field/both chat		
P3 mentions: field P 0.76 $\rightarrow$ 0.50 (neither 0.46); no mixed field to chat direction		
P4 erasure: A2 chain K -0.15 , idle -0.11 , work $\times$ browse up, $+0.11$; V4 K $+0.27$ $ K < 0.1$		
P5–P8 steps	unchanged (B4 states unaffected by as round the bin bug)	1
ITT (lever_design)	nudge work/chat share $+0.020$ [0.010, -0.030]	
Native NE43	nudger off: catalytic loss (B4, V4); mixed bookends off: B4 neither, V4 both	
Native NE44	12-h escape 0.14 vs 5-min 0.40	$\times$
Native NE10 (V4)	neither; wait share $+0.02$	mixed



Second figure. Round 1b. (a) The lever plane: round 1 (open), round 1b on B4 (filled) and on Jev v3.1 states (triangles); nudges stay in the “both” corner in both state spaces, erasure is a field with state-space-dependent catalysis. (b) Native steps inside #51 and at NE10: percentiles of idle/wait escape and occupancy among same-shape day-boundary placebos; the nudger switch-off sits at the bottom of the escape distribution with occupancy inside the band.

GOAL PERIODS IN DETAIL

G03 d 1b: descriptive (r1 descriptive)
G04 $\sim$ 1b: mixed (r1 supported)
G05 d 1b: descriptive (r1 descriptive)
G06 $\sim$ 1b: mixed (r1 mixed)
G08 $\times$ 1b: failed (r1 failed)
G10 $\times$ 1b: failed (r1 failed)
G11 $\times$ 1b: failed (r1 failed)
G12 $\times$ 1b: failed (r1 failed)
G13 $\sim$ 1b: mixed (r1 mixed)
G16 $\times$ 1b: failed (r1 failed)
+35 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The nudge class rests on one period (G51; G38 has the same signs but is not significant). Fields are invisible below ~ 100 episodes, so “catalyst” or “neither” in small periods does not exclude a field. The field/catalyst split depends on a load-distribution convention; $|K| < 0.10$ is not interpretable. B4 states mean different things in regimes I and III. Mentions are partly common-cause. Design choices made after seeing data are flagged: the no-consolidate erasure chain, the content-TV check, and the post hoc P9 switch-off. 28 class-level tests: the nudge field alone does not survive Bonferroni.

ROUND 2

Run `confirm.py` after commit (expect low nudge power on #45–#50). Agent-centered content states for steps; a dose analysis of nudger re-firing; the 08-21 switch-off as a proper NE with day-matched controls.

No round-2 redirects yet.

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